

WASP-36b

R-1609

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-36_210315 · created 2026-08-02T03:11:34+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459288.7289 +/- 0.0023 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.01
Transit depth (Rp/Rs)^2	2.61 +/- 0.32 [%]
Orbital Inclination (inc)	83.19 +/- 1.02
Airmass coefficient 1 (a1)	1.012 +/- 0.012
Airmass coefficient 2 (a2)	-0.009 +/- 0.014
Scatter in the residuals of the lightcurve fit is	1.19 %
Best Comparison Star	#1 - [339, 166]
Optimal Method	PSF photometry
Transit Duration (day)	0.0781 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.162 ± 0.01 vs 0.1368 (deviation 2.43σ)
Duration fit vs published	0.0781 d vs 0.0758 d (deviation 0.34σ)
Mid-time O–C	5.2 min
Depth SNR	15.6
Residual scatter	1.19 %

Light curve

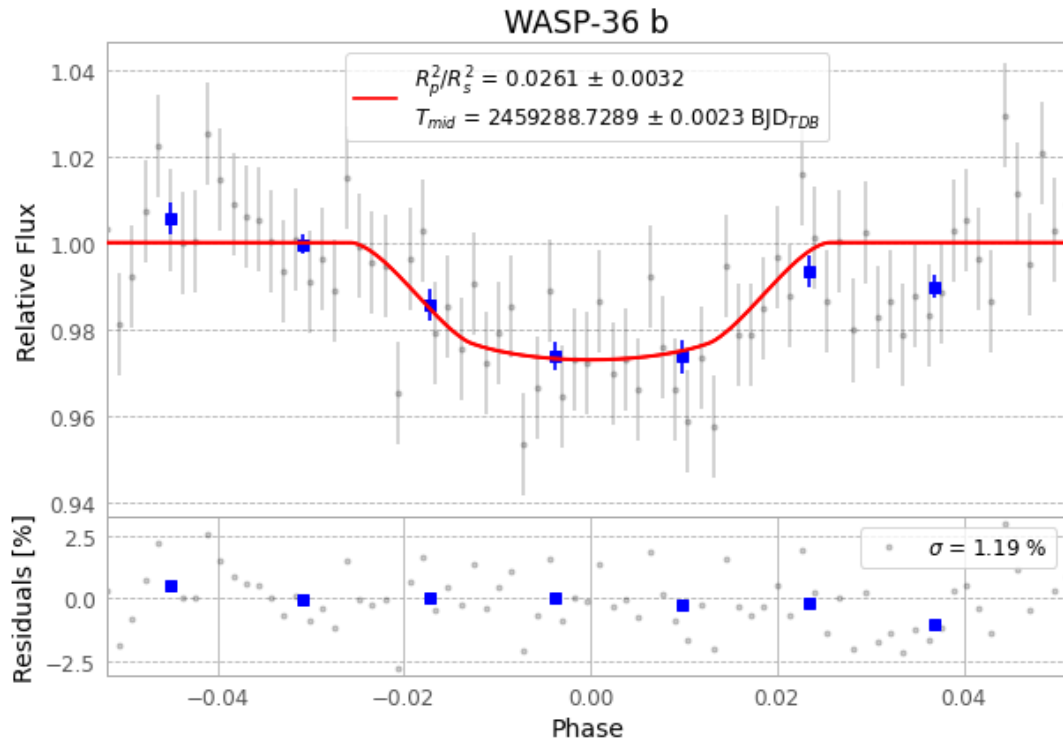


Figure 1 — FinalLightCurve_WASP-36 b_2021-03-14.png (SHA-256 23ce7c0e6cd843e6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [423, 213], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 af3d7a4f788ec8ed...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-36210315032714.FITS → WASP-36210315071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-210315.log (31e9b8b659fa...), FinalLightCurve_WASP-36 b_2021-03-14.png (23ce7c0e6cd8...), FinalLightCurve_WASP-36 b_2021-03-14.csv (603efacb7251...)

Compute resources

Wall time 150.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 03:11Z.